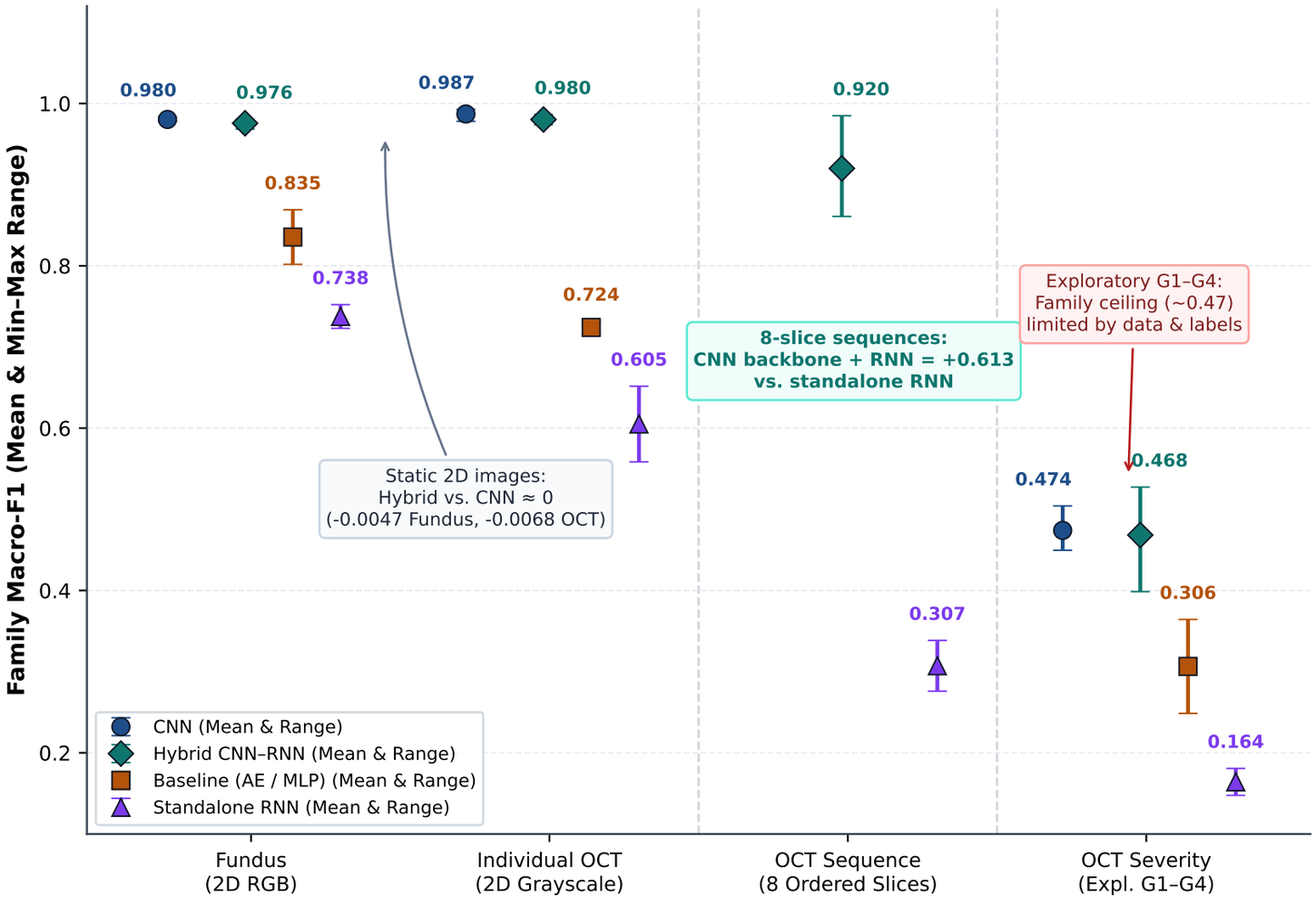

(b) Optimal Track Regimes, Cost & Methodological Caveats

Fundus 3-Class | Best: EfficientNet-B0

Macro-F1: 0.9835 ± 0.0055 • Params: 4.01M • Latency: 2.54 ms/img

Constraint: Near-duplicate clusters (n=205 test); no patient IDs;
100% of EfficientNet-B0 errors fall on Pseudopapilledema.

Individual OCT 3-Class | Best: DenseNet-121

Macro-F1: 0.9928 ± 0.0099 • Params: 6.95M • Latency: 3.89 ms/img

Constraint: Strongest 3-class separability, yet Normal & Pseudopapilledema
contain only 1 inferred subject prefix each.

OCT Sequence 3-Class | Best: ResNet-50 + BiLSTM

Macro-F1: 0.9849 ± 0.0338 • Params: 24.44M • Latency: 29.67 ms/win

Constraint: Hybrid CNN–RNN essential over RNN (+0.613), yet evaluated
on only 23 test windows across 4 inferred series.

OCT Severity (Exploratory G1–G4) | Best: ResNet-50 + LSTM

Macro-F1: 0.5274 ± 0.0780 • Params: 24.24M • Latency: 3.80 ms/img

Constraint: Exploratory folder-derived labels (unvalidated clinically);
severe G2 majority dominance & 96.9% G3 failure.